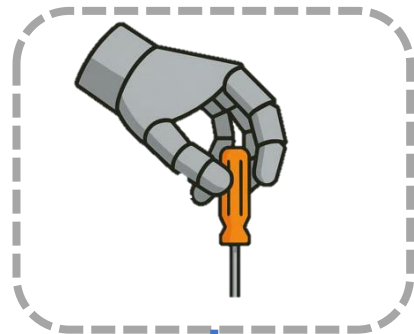
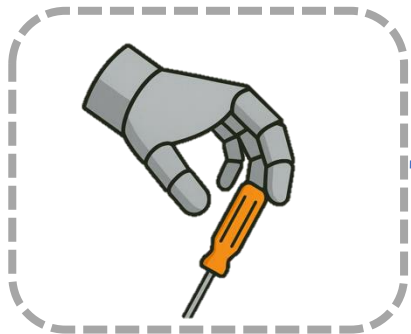


Primary manipulation

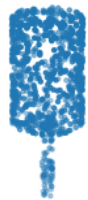


Drop detected ?

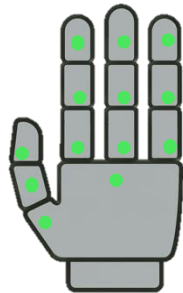


Yes

Object
geometry points



Predefined key
points



⋮

NDF
model

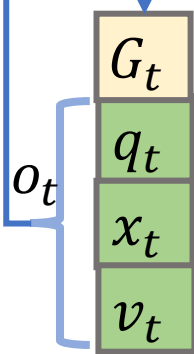


⋮

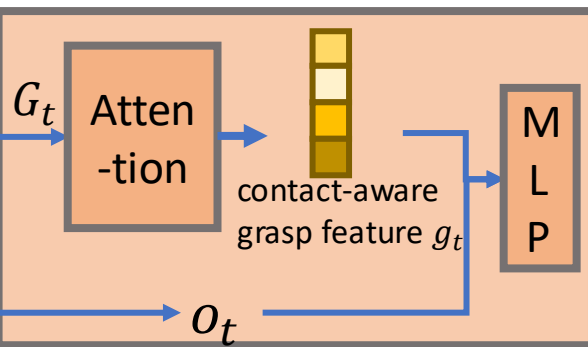


Raw grasp
feature G_t

Observations

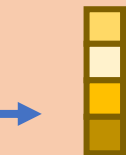


o_t



G_t

Atten-
tion



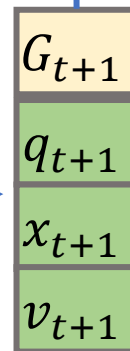
contact-aware
grasp feature g_t

M
L
P

a_t

Env

Reward r
+

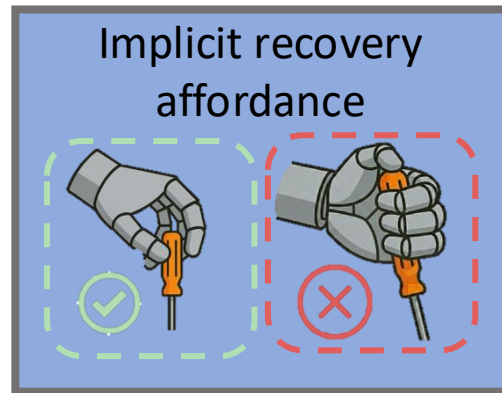


G_{t+1}

q_{t+1}

x_{t+1}

v_{t+1}



Implicit recovery
affordance

1
Train offline,
Freeze online

2
Online RL
training

Recovery
reward
 $r_{recover}$